

Roll No.

TCE-501

**B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

ENVIRONMENTAL ENGINEERING-I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Mention and discuss the factors that influence per capita demand. (CO1)

OR

- (b) Discuss the steps taken to prevent wastage and leakage of water in the distribution system. (CO1)

2. (a) Describe the peak hour demand and how it affects the design of the water supply system. (CO1)

OR

- (b) Describe the concept of a coincident draft and how it is used in estimating the design draft for the design of the distribution system. (CO1)

P. T. O.

3. (a) Assuming the geometric rate of growth of the population of a town, calculate the population of the town in the year 2021 with the help of the following census record of the population : (CO1)

Year	Population in Thousands
1991	300
2001	375
2011	440

OR

- (b) "The incremental increase method of estimating population combines the advantages of other 2 methods i.e. arithmetical increase and geometrical increase." Discuss the statement. (CO1)
4. (a) Water has to be supplied to a town population of 1,00,000 at the rate of 200 lpcd from a river, 1.8 kilometres away. The difference in elevation between the lowest hotel level in the sum and service reservoir is 36 metres. Determine the size of the main and the horsepower of the pump required. Assume the value of C is equal to 120 for the given pipe material and the efficiency of the pump set equal to 65%. (CO2)

OR

- (b) Describe the phenomenon of water hammer and how it is produced in pipes conveying water under pressure. Also, list out the precautions one should take to reduce its effect. (CO2)

(3)

5. (a) Distinguish between the following : (CO2)

- (i) Expansion joints and flanged joints
- (ii) Gravity conduits and pressure conduits

OR

(b) Write short notes on the following : (CO2)

- (i) Cement concrete and RCC water mains
- (ii) Corrosion of metal pipes